

1.

Let A be an $n \times m$ matrix whose entries belong to $\{+1, -1, 0\}$. Suppose that each column of A contains *exactly one* entry equal to $+1$ and *exactly one* entry equal to -1 .

Which of the following conclusions can be drawn?

A. $\text{Rank}(A) \leq n - 1$

B. $\text{Rank}(A) = m$

C. $n \leq m$

D. $n - 1 \leq m$

2.

Let A be a non-singular $n \times n$ matrix with real entries. Let $B = A^T$ denote the transpose of A .

Which of the following matrices is necessarily *positive definite*?

A. $A + B$

B. AB

C. $A^{-1} + B^{-1}$

D. ABA

3.

Minimum of dimension of the intersection of two seven dimensional vector subspaces in a twelve dimensional vector space is:

📁 Linear Algebra

#goclasses2026-da-aimt-2

#numerical-answers

#goclasses-testseries-2026

#linear-algebra

#one-mark

Let A be a linear transformation. Suppose

$\{v_1, w_1\}$ is a **linearly dependent** set of vectors and

$\{v_2, w_2\}$ is a **linearly independent** set of vectors, where

$$A(v_1) = v_2 \quad \text{and} \quad A(w_1) = w_2.$$

Which of the following statements is **correct**?

- A. Such a linear transformation A always exists
- B. Such a linear transformation A exists if $v_1 \neq w_1$
- C. Such a linear transformation A exists only if $v_2 = cw_2$ for some scalar c
- D. Such a linear transformation A can never exist

Let A and B be 2×2 matrices whose entries belong to $\{0, 1\}$.

All arithmetic is performed *modulo 2*, that is, $1 + 1 = 0$.

Suppose that $\text{tr}(A) = \text{tr}(B)$ and $\text{tr}(A^2) = \text{tr}(B^2)$.

Which of the following statements is/are necessarily true?

- A. A and B have the same eigenvalues
- B. A and B have the same determinant
- C. A and B may have different eigenvalues
- D. $A = B$



Let

0



$$A = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad Q(x) = x^T A x \quad \text{for } x \in \mathbb{R}^3.$$

Which of the following statements is/are **TRUE**?

- A. A has exactly two positive eigenvalues.
- B. All the eigenvalues of A are positive.
- C. $Q(x) \geq 0$ for all $x \in \mathbb{R}^3$.
- D. $Q(x) < 0$ for some $x \in \mathbb{R}^3$.



1



For any $n \times n$ matrix B , let $N(B) = \{x \in \mathbb{R}^n \mid Bx = 0\}$ denote the null space of B .

Let A be a 4×4 matrix such that

$$\begin{aligned} \dim(N(A - 2I)) &= 2, \\ \dim(N(A - 4I)) &= 1. \end{aligned}$$

and $\text{rank}(A) = 3$.

Then, which of the following statements is/are true?

- A. 0, 2, and 4 are eigenvalues of A .
- B. $\det(A) = 0$.
- C. A is not diagonalizable.
- D. $\text{trace}(A) = 8$.

Let

$$S = \left\{ (x_1, x_2, x_3, \dots, x_{50}) \in \mathbb{R}^{50} \mid \begin{array}{l} x_1 = x_2 = \dots = x_{25} = 0, \\ x_{26} + x_{27} + \dots + x_{50} = 0 \end{array} \right\}.$$

The dimension of the vector space S is:

Let S and T be vector subspaces of a vector space V . Then $S \cup T$ is a subspace of V

- A. is never true
- B. if and only if one of S or T is trivial
- C. if and only if $S \subseteq T$ or $T \subseteq S$
- D. if and only if $S \cap T$ is a nonzero vector space.

Suppose A is a 3×3 symmetric matrix such that

$$[x, y, 1]A \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = xy - 1$$

Let p be the number of positive eigenvalues of A and let $q = \text{rank}(A) - p$. Then

A. $p = 1$.

B. $p = 2$.

C. $q = 2$.

D. $q = 1$.

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

and let I denote the 3×3 identity matrix.

If $6A^{-1} = aA^2 + bA + cI$, for real numbers $a, b, c \in \mathbb{R}$, then the ordered triple (a, b, c) is equal to:

A. $(1, 2, 1)$

B. $(4, 1, 1)$

C. $(1, -1, 2)$

D. $(1, 4, 1)$

Let A be an $m \times n$ real matrix such that AA^T is invertible. Which of the following statements is/are necessarily true?

A. $\text{rank}(A) = m$

B. $\text{rank}(A) = n$

C. $m = n$

D. A is invertible

Consider a real matrix A of size $m \times n$.

Which of the following statements is/are necessarily true?

A. If the row space of A is $\mathbb{R}^n$, then the column space of A is $\mathbb{R}^n$.

B. If the null space of A is $\mathbb{R}^n$, then the column space of A is the zero vector space $\{0\}$.

C. If the left null space of A is $\mathbb{R}^m$, then the column space of A is the zero vector space $\{0\}$.

D. If the row space of A is $\mathbb{R}^n$, then the column space of A is $\mathbb{R}^m$.

Consider a matrix A such that its null space $\mathcal{N}(A)$ is spanned by a nonzero vector v . Which of the following matrices represents the orthogonal projection onto the row space of A ?

A. $\frac{vv^T}{v^T v}$

B. $I - \frac{vv^T}{v^T v}$

C. $\frac{v^T v}{vv^T}$

D. $I + \frac{vv^T}{v^T v}$

Consider the following problem.

Let A be a block matrix with real entries, with each block B_{ij} of size 2×2 , given by

$$A = \begin{bmatrix} B_{11} & B_{12} & B_{13} & B_{14} \\ 0 & B_{22} & B_{23} & B_{24} \\ 0 & 0 & B_{33} & B_{34} \\ 0 & 0 & 0 & B_{44} \end{bmatrix},$$

where

$$B_{11} = \begin{bmatrix} 2022 & 2021 \\ 2021 & 2022 \end{bmatrix}.$$

Which of the following statements is correct?

- A. $\lambda = 1$ is an eigenvalue of A
- B. $\lambda = 1$ is not an eigenvalue of A
- C. $\lambda = 0$ is the only eigenvalue of A
- D. All eigenvalues of A are strictly greater than 1

Given a matrix $\mathbf{X}$ of shape $D \times N$ with a singular value decomposition (SVD), $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\top$, suppose $\mathbf{X}$ has rank K and $\mathbf{A} = \mathbf{X}\mathbf{X}^\top$

Which one of the following statements is/are true?

- A. The eigenvalues of $\mathbf{A}$ are the singular values of $\mathbf{X}$
- B. $\mathbf{A}$ is positive semi-definite, i.e all eigenvalues of $\mathbf{A}$ are non-negative
- C. The eigendecomposition of $\mathbf{A}$ is also a singular value decomposition (SVD) of $\mathbf{A}$
- D. A vector $\mathbf{x}$ that can be expressed as a linear combination of the last $D - K$ columns of $\mathbf{U}$, i.e $\mathbf{x} = \sum_{i=K+1}^D \mathbf{w}_i \mathbf{u}_i$ (where $\mathbf{u}_i$ is the i -th column of $\mathbf{U}$), lies in the null space of $\mathbf{X}^\top$

Let $u_1, u_2, u_3, v_1, v_2, v_3$ be vectors in $\mathbb{R}^4$. Let U be the span of $\{u_1, u_2, u_3\}$ and let V be the span of $\{v_1, v_2, v_3\}$.

Consider the following statements:

- I. If the dimension of $U \cap V$ is 2 and the dimension of U is 3, then $\{v_1, v_2, v_3\}$ is linearly dependent.
- II. If $U + V = \{u + v : u \in U, v \in V\} = \mathbb{R}^4$, then either $\{u_1, u_2, u_3\}$ is linearly independent or $\{v_1, v_2, v_3\}$ is linearly independent.

Which of the above statements is/are true?

- A. Only I
- B. Only II
- C. Both I and II
- D. Neither I nor II

Let A and B be real, invertible matrices such that

$$AB = -BA$$

Then which of the following statements is correct?

(Here $\text{Tr}(\cdot)$ denotes the trace of a matrix.)

- A. $\text{Tr}(A) = \text{Tr}(B) = 0$
- B. $\text{Tr}(A) = \text{Tr}(B) = 1$
- C. $\text{Tr}(A) = 0, \text{Tr}(B) = 1$
- D. $\text{Tr}(A) = 1, \text{Tr}(B) = 0$

Let A be a *real symmetric matrix* whose eigenvalues are $\lambda_1, \lambda_2, \dots, \lambda_n$.

Define a polynomial by

$$p(t) = 1 + t.$$

For any vector x , let $\|x\|$ denote its **Euclidean length**.

Determine the value of

$$\max_{\|x\|=1} \|p(A)x\|.$$

- A. $\max\{1 + \lambda_1, 1 + \lambda_2, \dots, 1 + \lambda_n\}$
- B. $\max\{|1 + \lambda_1|, |1 + \lambda_2|, \dots, |1 + \lambda_n|\}$
- C. $\min\{1 + \lambda_1, 1 + \lambda_2, \dots, 1 + \lambda_n\}$
- D. $1 + \lambda_1 + \lambda_2 + \dots + \lambda_n$

Let A be an $m \times n$ real matrix and consider the system $Ax = b$.

Which of the following statements are **correct**?

- A. Any solution x of $Ax = b$ (if it exists) can be expressed as the sum of a vector from the **null space of A** and a vector from the **row space of A** .
- B. The system $Ax = b$ is solvable if and only if b is orthogonal to every vector in the **null space of A** .
- C. If A is a 4×3 matrix and the solutions of $Ax = b$ are not unique whenever they exist, then the rank of A must be exactly 2.
- D. If x_1 and x_2 are two solutions of $Ax = b$, then $x_1 - x_2$ lies in the **null space of A** .